Experiment 7

API INTEGRATION WITH ZOHO CREATOR

Aim

To integrate Zoho Creator with external applications and services using REST APIs to enable secure data exchange, automate administrative workflow processes, and extend the functionality of the **Application Form Management System**.

Theory

API integration allows Zoho Creator to communicate and exchange data seamlessly with external systems. In the Application Form system, REST APIs are used to retrieve application records, transmit submission metrics to analytics tools, verify fee payments, and interface with external verification services via OAuth authentication and standard Deluge scripts using `invokeUrl`.

Procedure

1. Log in to Zoho Creator and open the **Application Form Management System** workspace.
2. Identify the external target service (e.g., Zoho Analytics or the `Full_Applications_Report` REST API).
3. Configure API access and authentication settings using an OAuth 2.0 connection.
4. Set up necessary API credentials, client keys, and access scopes (e.g., Zoho`Creator.reports.READ`).
5. Write Deluge integration scripts to send or receive data from the **Application Form**.
6. Configure request URLs, HTTP methods (GET/POST), parameters, and payload structures.
7. Test the integration and verify that application records are fetched and transmitted correctly.
8. Implement error-handling routines to capture status codes and validate returned JSON responses.

Applications of API Integration

- **Analytics Integration:** Transmitting Application Form data to external BI dashboards for real-time submission metrics.
- **Master Data Sync:** Retrieving institution/department lookup codes from external master repositories.
- **Automated Notifications:** Connecting third-party SMS/Email gateways for submission acknowledgments.
- **Payment Verification:** Verifying application fee transactions with external gateway endpoints.

API Request Specification

Configuration Item	Details / Value
HTTP Method	`GET`
Target Endpoint	`Full_Applications_Report` via Zoho Creator REST API v2
Authentication	OAuth 2.0 connection with read scope (Zoho`Creator.reports.READ`)
Purpose	Fetching submitted application records for analytics, review, and reporting

Sample Deluge Integration Script

```
baseUrl = "https://creator.zoho.in/api/v2/<account_owner>/";
response = invokeUrl
[
    url: baseUrl + "application-form-management/report/Full_Applications_Report"
    type: GET
    connection: "creator_connection"
];

info response;

if (response.get("code") == 3000)
{
    records = response.get("data");
    info "Applications fetched: " + records.size();
}
else
{
    info "API error response: " + response;
}
```

Expected Response Format (JSON)

```
{
  "code": 3000,
  "data": [
    {
      "Application_ID": "APP-2026-001204",
      "Applicant_Name": "Ananya Ramesh",
      "Selected_Category": "General / B.Tech Computer Science",
      "Net_Application_Fee_Paid": 4250
    }
  ]
}
```

Integration Architecture Flow



Error Handling & Best Practices

- **Status Validation:** Check `response.get("code") == 3000` before processing dataset contents.
- **Audit Logging:** Log unexpected response codes to notify administrators of unexpected service outages.
- **Credential Safety:** Keep tokens stored securely inside OAuth Connection objects; avoid hardcoding secrets in scripts.
- **Query Optimization:** Fetch only necessary fields and records to maintain fast API response cycles.

Result & Conclusion

The **Application Form System** was successfully integrated conceptually with the Zoho Creator REST API and external services. API integration enables seamless data exchange, real-time analytics, and automated processing while ensuring full security and error tolerance.